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REGULAR PAPER

Decoding reddit memes virality

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Abstract

Understanding the elements that contribute to a meme's virality is crucial due to the significant role memes play in digital culture and communication. This study analyzed 16,968 memes from Reddit, which are now available on the Open Science Framework. It aimed to identify predictors of virality, focusing on both textual and image-based characteristics. Our findings indicate that memes featuring multiple objects, humans, or animals, and those utilizing less saturated colors such as light gray and dark gray, tend to achieve higher virality. Additionally, smaller thumbnail dimensions and the avoidance of surprise facial expressions are associated with higher virality. Notably, memes posted between 12 pm and 6 pm UTC show a significant increase in virality. Combining these predictors enhances the area under the receiver operating characteristic curve of virality predictions, highlighting the complex interplay between content elements in meme dissemination. This research underscores the importance of a multifaceted approach in predicting meme virality and offers insights into effective social media strategies. Additionally, we provide an open-source codebase on GitHub to promote research transparency and reproducibility.

Keywords Reddit memes virality prediction · Natural language processing · Computer vision · Sentiment analysis · Feature importance · Object detection

1 Introduction

The term "meme" was originally introduced by Richard Dawkins in his 1976 book "The Selfish Gene" to denote an idea, behavior, or style that spreads from person to person within a culture [1]. As per Davison's contemporary definition, "An Internet meme is a piece of culture, typically a joke, which gains influence through online transmission" [2]. Internet memes have gained immense popularity, captivating the attention of people. For example, Figs. 1 and 2 represent viral memes, while Figs. 3 and 4 depict non-viral memes, illustrating the contrast between the two categories. Throughout this paper, the term 'memes' specifically refers to internet memes. However, not all memes achieve viral status, prompting the question: can we accurately predict which memes will go viral? This research into predicting viral memes holds intrigue not only for meme creators and enthusiasts but also carries significance for marketers, advertisers, and social media platforms. Businesses often leverage the virality of memes as a potent tool for online promotion and engagement. Understanding the patterns leading to meme popularity enables content creators and marketing strategists to tailor their approaches and enhance the likelihood of their content going viral. While some studies have attempted to predict the virality of images, limited work is done to predict the virality of memes, particularly on platforms like Reddit.

The importance of understanding meme virality extends beyond entertainment. Memes play a pivotal role in shaping online discourse, influencing cultural trends, and even driving public opinion. This research provides actionable insights for marketers, content creators, and social media platforms, enabling them to optimize their content strategies and amplify audience engagement. By focusing on Reddit, a platform with diverse and community-driven meme culture, this study addresses a critical gap in existing research, offering unique perspectives on meme dynamics.

Furthermore, the implications of this study are not limited to content creation. Predicting meme virality can help identify and mitigate the spread of harmful or misleading content, making online platforms safer and more informative. The interdisciplinary nature of this research—integrating computer vision, natural language processing, and tempo-

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ral analysis—also lays the groundwork for advancements in AI-driven social media analytics. These contributions pave the way for cross-platform studies and deeper insights into the mechanisms of digital content virality.

To address this gap, our study makes several contributions to the understanding of meme virality on Reddit:

- **Provides an open-source codebase on GitHub.** This promotes research transparency and reproducibility.
- **Investigates how visual elements affect meme popularity on Reddit.** Focuses on color, composition, and imagery.
- **Analyzes textual content's impact on user engagement and virality.** Focuses on texts within memes and subreddit titles.
- **Assesses how upload times influence meme popularity.** Identifies peak posting intervals for maximum engagement.
- **Compares textual and image-based features for predicting virality.** Provides insights into content effectiveness.
- **Tests a combined model of textual and image-based features.** Aims for better prediction accuracy of meme virality.
- **Prepares for cross-platform studies on meme virality dynamics.** Sets a precedent for analyses across various social media.
- **Equips content creators with strategies for better engagement.** Enhances organic reach by decoding virality patterns.
- **Makes available a dataset of 16,968 Reddit memes, stored on the Open Science Framework (OSF).** Enables further research and verification of our findings, fostering deeper insights into meme virality.

Fig. 1 Viral meme with an average saturation of 45 and an average value of 167, both on a 0–255 scale. The meme's sentiment is positive in both the text embedded in the meme and the meme title, and it features both an animal and a human

Fig. 2 Viral meme with an average saturation of 31 and an average value of 197, both on a 0–255 scale. It features two humans and was uploaded in the afternoon (12–3 PM UTC)

2 Related work

2.1 Visual features and content popularity

Research on the predictors of online content popularity has examined visual characteristics as well as user interaction metrics. Khosla et al. [3] researched various low-level visual features such as color and texture. They also explored high-level content attributes like object presence and social context on Flickr. Their study demonstrated how these factors correlate with content popularity.Similarly, Aloufi et al. [4] conducted research on similar aspects. Mazloom et al. [5] extended this research to Instagram, identifying that training a popularity model on specific categories like action, scene, people, and animals increases the accuracy compared to training a model on all data.

2.2 Textual content and user demographics

Extending beyond just visual features, the influence of textual content and user demographics on the popularity and sharing behavior of online media has also been significant. Du et al. [6] analyzed the impact of textual content in visual memes on Twitter, relating it to user demographics and meme-sharing behaviors. They found that certain demographics are more likely to express themselves through Image-with-text (IWT) memes, as compared to memes without text, images in general, and text-only tweets. This multidimensional approach not only underscores the complexity of content virality but also highlights the broad spectrum of factors that predictive models must consider.

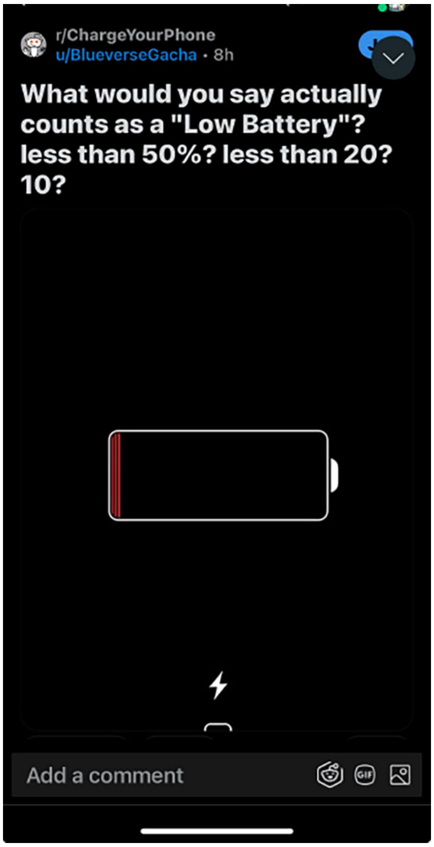


Fig. 3 Non-viral meme with an average saturation of 7 and an average value of 11, both on a 0–255 scale. It was uploaded in the morning (6 AM–9 AM UTC) and included the submission text

Fig. 4 Non-viral meme with an average saturation of 97 and an average value of 158, both on a 0–255 scale. It included the submission text, and the meme title has a negative sentiment

2.3 Psychological and sociological aspects of memes

The psychological and sociological impacts of memes have garnered extensive focus, particularly regarding how they resonate and proliferate through communities. Miltner [7] explored the use of memes in political and activist contexts, charting their evolution from grassroots tools to commercialized elements within mainstream media. This transition underscores memes' dual role in engaging public interest and serving commercial ends. Following this, Nieubuurt [8] provided an insightful analysis of memes' ability to capture attention through dense modalities and exploitation of cognitive biases, likening them to modern-day leaflet propaganda that navigates the information overload of the internet age effectively.Further examining the psychological effects, Das [9] applied Social Cognitive Theory and Uses and Gratifications Theory. The study explored how memes influence emotional resonance and amplify social echo chambers. These effects, in turn, impact public opinion and social discourse. On an evolutionary scale, Blackmore [10] posited that memes are fundamental cultural units that have co-evolved with human genes, playing a significant role in shaping human evolution.

2.4 Meme diffusion models and dynamics

Understanding how memes spread has been another focus of research. Johann [11] utilized Spitzberg's model of meme diffusion and Rogers' diffusion of innovations framework to analyze the spread of image-based memes on platforms like Twitter. His quantitative analysis of the Merkel Meme identified critical factors for viral diffusion, such as the involvement of early adopters and the adaptability of meme content through image editing. Complementing this, Leiser and BULOW [12] investigated the motivations behind meme consumption and sharing within political contexts, revealing that individuals engage with memes for reasons spanning self-expression, entertainment, and social identity affirmation

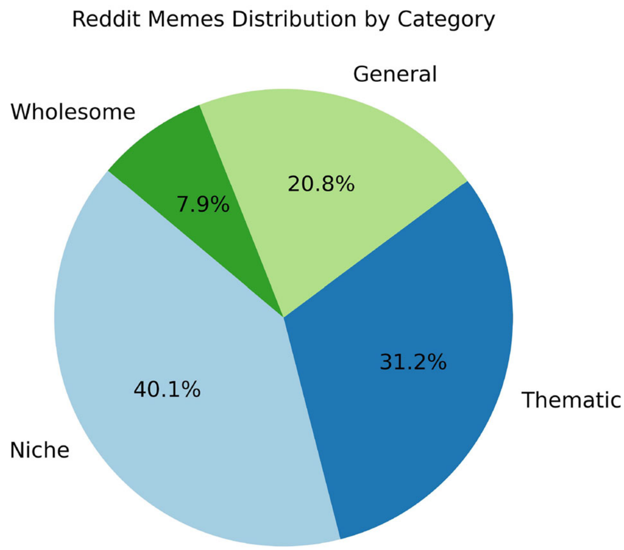


Fig. 5 Reddit memes distribution by category: general, thematic, wholesome, and niche content. This chart highlights the proportional representation of posts across these categories

2.5 Technical advancements in predicting content popularity

In terms of technical advancements, the application of machine learning to predict content popularity has shown substantial promise. Gelli et al. [13] harnessed these techniques to predict the popularity of social images on Flickr by using visual features combined with user-related data, illustrating the potential of machine learning to decode complex patterns from large datasets. Building on different methodologies, Yang et al. [14] introduced a blockchain-based model for simulating meme propagation, and Maji et al. [15] utilized social network analysis to study meme dynamics within online communities. Additionally, Ling et al. [16] focused on the specific features that contribute to the virality of image memes, although their analysis was limited by the small size of their dataset and relied primarily on descriptive statistics to understand the impact of various variables. Similarly, Barnes et al. [17] predicted meme virality based solely on content; however, their analysis was specifically conducted during the COVID-19 pandemic. Overall, while existing research sheds light on various aspects of online content popularity, there remains a gap in comprehensive studies specifically focused on predicting the virality of Reddit memes.

2.6 Identifying gaps and our contribution

While prior studies provide valuable insights into visual, textual, psychological, and technical aspects of content popularity, a comprehensive analysis focused on predicting the virality of Reddit memes is lacking.To bridge this gap, we conducted research to identify the statistically significant textual features, image features, and upload times associated with predicting virality. Additionally, we assessed the relative importance of features extracted from memes in both textual and image domains, as well as the combined importance of these features. Furthermore, through univariate logistic analysis, we explored the direction and magnitude of these features in predicting virality. We delved into whether textual features outperform image features in predicting virality or vice versa and whether a combination of these features is necessary for accurate virality prediction with the assistance of the CatBoost algorithm [18].

3 Data and methods

3.1 Dataset extraction

In January 2024, our study extracted 16,968 posts from 29 subreddits selected for their engagement with meme culture. These subreddits were identified through curated lists (e.g., Reddit's Best Communities, RedditList) and targeted searches. Targeted searches systematically explored Reddit's search functionality to identify meme-related subreddits with high activity levels and community engagement. Keywords such as meme, funny, humor, and wholesome were used to locate these subreddits.

To ensure diverse representation of meme culture, subreddit descriptions, top posts, and recent activity were manually reviewed to identify communities specializing in distinct meme genres or unique humor styles. Subreddits such as HistoryMemes and PoliticalHumor were included for their thematic focus, while dankmemes and me_irl were selected as general hubs for meme creation and sharing. Communities like surrealmemes and bonehurtingjuice captured niche meme formats, ensuring a broad spectrum of meme culture. Subreddits like wholesomememes represented uplifting, feel-good content. The dataset's category distribution—General (20.8%), Thematic (31.2%), Wholesome (7.9%), and Niche (40.1%)—is illustrated in Fig. 5.

However, several subreddits were not included. For example, MemeEconomy was excluded because it primarily focuses on the discussion and "trading" of meme trends rather than direct meme sharing, making its content less suitable for our image-based analyses. Similarly, dankchristianmemes was omitted due to its highly niche, religion-centered memes, which limited its generalizability to broader meme culture. Additionally, MemeTemplatesOfficial was not included as it serves as a repository for meme templates rather than a community sharing finished memes, thus lacking the user engagement patterns we wished to study. By excluding such highly specialized or discussion-focused subreddits, we aimed to concentrate on communities actively sharing and engaging with finished meme content in a broad range of styles.

This variety in subreddit categories was one of three key criteria guiding subreddit selection, along with subscriber count and engagement. Popular subreddits such as memes (29.4M subscribers) and dankmemes (6 M) were included for their influence, while niche subreddits like DeepFried (18.5K subscribers) and bonehurtingjuice (903K subscribers) ensured the inclusion of less mainstream but culturally significant formats. Engagement metrics, measured through mean upvotes, revealed high interaction in subreddits like WhitePeopleTwitter and blackpeopletwitter (over 3,000 upvotes per post), as shown in Fig. 6. Niche subreddits like DeepFried demonstrated engagement proportional to their size, reflecting active community participation.

Together, these criteria ensure that the selected subreddits represent a balanced mix of popular and niche communities, covering dominant and emerging trends in meme culture. While it is impractical to analyze all meme-related subreddits on Reddit, the selected 29 subreddits account for a substantial portion of meme-related activity and diversity, making them representative of Reddit's broader meme ecosystem.

We extracted the data using the Python Reddit API Wrapper(PRAW) library, leveraging authenticated access provided by `client_id`, `client_secret`, and `user_agent` [19]. These credentials ensured compliance with Reddit's API rate limits and security policies, critical for maintaining the integrity and adhering to ethical considerations of our data collection. Using PRAW, we accessed up to 1,000 posts from each subreddit in the 'hot' category, although the actual number retrieved was often less than this limit due to our stringent content filtering. Only posts with images in `.jpg`, `.jpeg`, or `.png` formats were collected to maintain relevance to our meme-focused research. Additionally, essential metadata such as titles, upvotes, comments, submission times, and image URLs were gathered. Table 6 in Appendix A presents a breakdown of subreddits, the number of memes, and corresponding subscriber counts. The first 19 rows in Table 7 in Appendix B display the initial features extracted from these posts. Posts with images smaller than ten pixels or displaying errors during processing were excluded to ensure data quality and accuracy. Following this filtering process, a dataset comprising 16,961 posts was obtained, and the resulting breakdown is documented in Table 6 in Appendix A.

The data was then securely stored in Datalore storage, with the complete dataset subsequently made publicly available on the Open Science Framework (OSF) to support transparency and facilitate further research in the field. This commitment to data accessibility is further exemplified by the public hosting of our extraction code on GitHub, which allows other researchers to replicate or build upon our methodology.

3.2 Dataset preprocessing

In the data preprocessing phase of our research, we feature-engineered a comprehensive set of features, categorized into three main types: image-related, text-related, and upload time features.

For image-related features, we employed YOLOv5 to detect various objects within the memes [20]. These objects were classified into 18 distinct categories, including humans, animals, and others, alongside a count and indicator for each detected object. This aligns with Social Cognitive Theory, which emphasizes the role of attention in shaping engagement. Visual elements such as the presence of humans or animals are likely to draw attention and contribute to a meme's ability to capture users' interest. We also used the facial expression recognition (FER) library to identify facial expressions depicted in the memes, classifying them into emotions such as happiness, anger, disgust, fear, neutrality, sadness, and surprise, with all remaining expressions labeled as unknown [21]. Emotions depicted in facial expressions also align with Social Cognitive Theory, as they reinforce retention and engagement. Color attributes were quantified by calculating the average hue, saturation, and value, and we marked the presence or absence of the top 10 colors identified in each meme.

Text-related features were extracted from memes using OpenCV [22] and EasyOCR [23]. Sentiment analysis was performed on the extracted text using the Hugging Face Transformers library [24]. The DistilBERT model was used to categorize sentiments into negative, neutral, and positive. Positive sentiment aligns with Uses & Gratifications Theory, reflecting the user's motivation to consume uplifting or entertaining content. We marked the absence of text as 'unknown' [25]. Moreover, we developed a 'call-to-action' indicator to assess whether memes encouraged specific user actions such as commenting, sharing, or following, and whether the presence of such calls-to-action contributes to their virality. This indicator was derived from the textual content embedded in memes and analyzed through a word cloud, as shown in Fig. 7. The word cloud highlights frequently used phrases associated with calls-to-action, such as 'like,' 'share,' 'comment,' and 'follow.' These terms were flagged as Boolean indicators of a 'call-to-action,' reflecting Uses & Gratifications Theory, which emphasizes users' motivation to engage with content that explicitly prompts interaction. We also analyzed subreddit post titles, measuring word length and sentiment with DistilBERT.

To examine the influence of posting times on meme popularity, we transformed UTC timestamps into descriptive daily segments, such as morning and night. This approach is grounded in Social Cognitive Theory, where the timing of content availability serves as a form of contextual reinforcement, shaping user engagement behaviors. By identifying

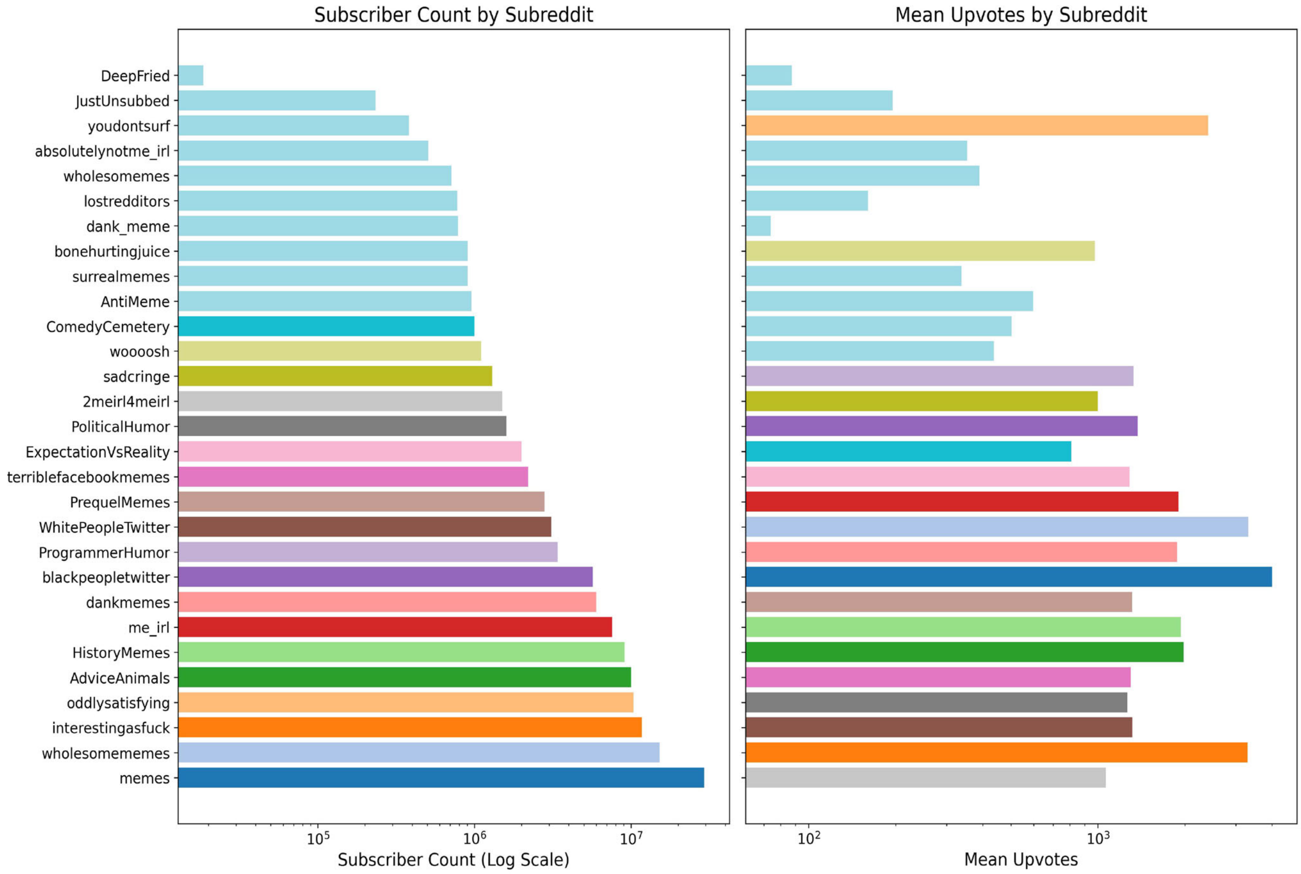
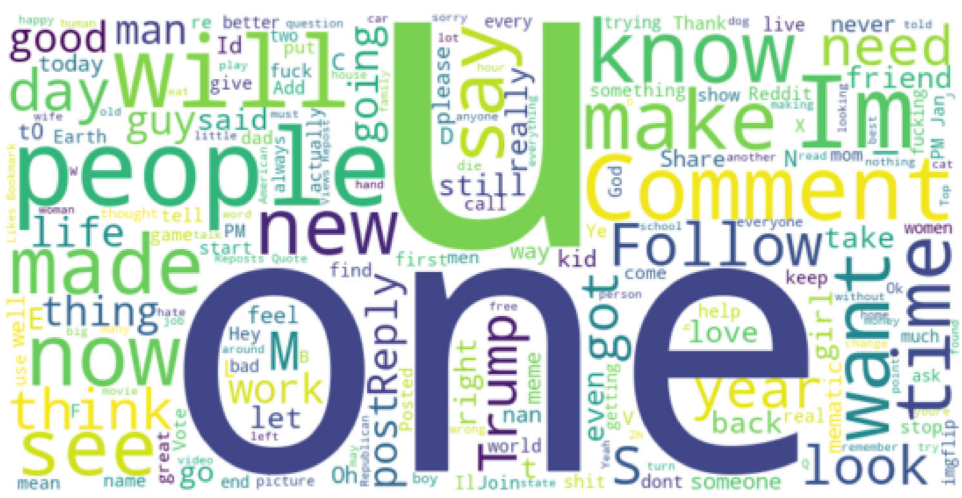


Fig. 6 Comparison of subscriber count and mean upvotes across subreddits

Fig. 7 Word cloud visualization of extracted text from memes



peak engagement periods, this analysis reveals optimal posting times to maximize visibility and virality.

3.3 What is reddit meme virality?

Virality on Reddit can be assessed using metrics such as upvotes, the ratio of upvotes to total votes, or the number of comments, each capturing different aspects of audience engagement. Initially, we chose upvotes as a measure of popularity because they directly reflect community approval and are easily accessible within Reddit's data. However, this approach introduced a bias tied to subreddit size. As shown in Fig. 6, subscriber counts strongly correlate with mean upvotes. The Pearson test (p-value = 0.09 and correla-

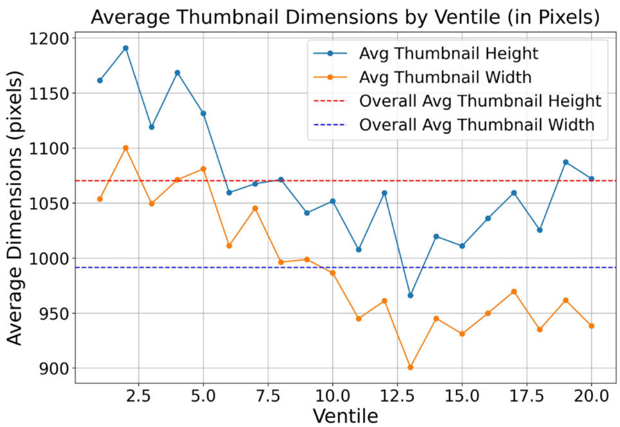


Fig. 8 Average thumbnail height and width across ventiles

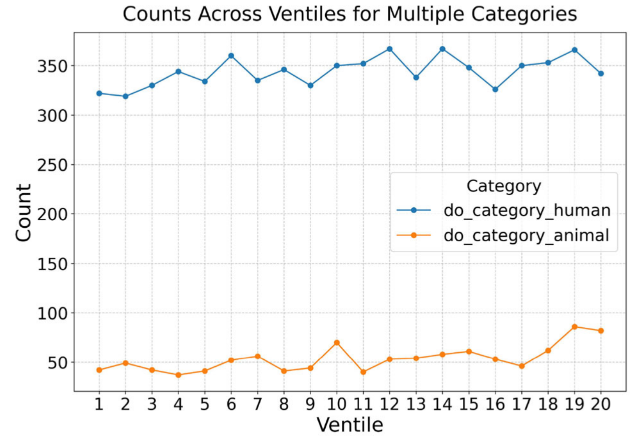


Fig. 9 Presence of animal count and presence of human count across ventiles

tion coefficient = 0.31) shows a slight correlation, while the Spearman test (p-value = 0.001 and correlation coefficient = 0.65) indicates a significant correlation.

To mitigate this bias, we subtracted the mean upvotes of the subreddit from the post's upvotes and then divided this difference by the range of upvotes (maximum minus minimum) within that subreddit. After normalization, correlation tests revealed no significant relationships (Pearson: p-value = 0.42, correlation coefficient = −0.15; Spearman: p-value = 0.62, correlation coefficient = 0.01). This adjustment provides a more robust and unbiased measure of virality, allowing us to account for variations in subreddit size and engagement patterns. Normalized metrics not only level the playing field but also offer a clearer evaluation of a meme's true potential to go viral on Reddit.

To differentiate viral from non-viral content, we designated posts at or above the 95th percentile as viral and those at or below the 5th percentile as non-viral. The percentiles are determined based on the upvote distribution of the entire dataset after normalization. This distinction allows us to focus on the most extreme and informative cases, avoiding the dilution of insights by including middle-range posts (5th-95th percentile).

Our analyses included continuous variables like Thumbnail Height and Thumbnail Width across 20 equally sized groups (ventiles), where each ventile represents $\frac{1}{20}$th (or 5%) of the sorted dataset based on normalized upvotes (Fig. 8) and categorical variables such as the count of human and animal presence across 20 ventiles (Fig. 9). These analyses revealed mixed signals across ventiles, further emphasizing the importance of isolating the extremes for a clearer understanding of the factors that influence virality.

Additionally, the choice of percentiles was carefully considered to strike a balance between capturing meaningful extremes and ensuring a sufficient sample size. While using the 1st and 99th percentiles might provide sharper distinctions, it would leave us with only 340 samples, limiting statistical power. On the other hand, the 10th and 90th percentiles would include too many posts, potentially diluting the distinct characteristics of viral and non-viral content. The 5th and 95th percentiles, therefore, represent a practical compromise capturing significant extremes while maintaining a robust sample size for meaningful analysis.

By defining virality in this manner, we get 1700 posts. Finally, all the extracted and engineered features from Reddit are listed in Table 7 in Appendix B.

3.4 Methods

In our study, we rigorously analyzed factors contributing to meme virality using a blend of statistical and machine learning techniques. We divided our dataset into training and testing sets with an 80-20 ratio, enhancing reproducibility by setting the random seed to 42. This split ensured robust model training and allowed for an unbiased evaluation of model performance.

Using the `statsmodels` library, we conducted logistic regression analyses to evaluate the impact of individual variables on meme virality, with the dependent variable encoded as 1 for viral and 0 otherwise [26]. Each variable was analyzed separately to calculate regression coefficients and p-values. Derived from the training set, these metrics provide a statistical basis for quantifying each variable's independent effect on the likelihood of a meme becoming viral, aiding in comprehensive feature evaluation.

For a more nuanced understanding of feature importance, we employed the CatBoost model, renowned for its effectiveness in handling categorical data and providing interpretable outputs [27]. Calculations of feature importance were based exclusively on the training set, identifying attributes that significantly influence the model's predictions. This approach not only facilitated the identification of key drivers of viral-

Table 1 Statistical analysis of visual features in memes: coefficients and p-values

Feature	Coefficient	p-Value
Presence of objects	0.2419	0.0300
Presence of multiple objects	0.4282	0.0000
Presence of single objects	−0.2773	0.0300
Facial expressions surprise	−1.0188	0.0100
Presence of humans	0.2177	0.0400
Presence of animals	0.4950	0.0000
Average saturation	−0.0035	0.0100
Average value	0.0045	0.0000
Thumbnail height	−0.0002	0.0400
Thumbnail width	−0.0003	0.0000
Dark slate gray	−0.3106	0.0400
Dim gray	−0.4799	0.0000
Dark gray	0.3486	0.0300
Black	−0.6462	0.0000
Light gray	0.7152	0.0100

ity but also helped enhance the predictive performance of our models.

To construct our predictive model, we explored various advanced machine learning algorithms, including CatBoost, LightGBM [28], and XGBoost [29]. Through a comparative analysis utilizing the Area Under the Curve (AUC-ROC) metric, CatBoost emerged as the superior model, delivering the highest AUC-ROC score, which was notably calculated based on the testing dataset.

Furthermore, to rigorously evaluate the predictive performance and significance of the AUC-ROC scores derived from different sets of features, we implemented DeLong's test [30]. This statistical test was applied to compare the AUC-ROC scores obtained from models using Image-Related Features versus Text-Related Features, Combined Features versus Text-Related Features, and Combined Features versus Image-Related Features. The application of DeLong's test allowed us to quantitatively assess the statistical significance of the differences in AUC-ROC scores, providing a robust statistical foundation for our findings on the relative importance of text, image, and combined features in predicting meme virality.

4 Results

This section outlines the findings from our comprehensive research across five distinct areas related to meme virality. Each subsection focuses on different factors contributing to a meme's popularity.

Table 2 Statistical analysis of textual features in memes and reddit posts: coefficients and p-values

Feature description	Coefficient	p-Value
Presence of submission text	−0.9574	0.0000
Positive sentiment in reddit post title	0.9084	0.0000
Negative sentiment in reddit post title	−0.9084	0.0000
Call-to-action indicator in memes	−0.3872	0.0016
Positive sentiment in meme text	0.3240	0.0180
Reddit post title word count	−0.0238	0.0138

Table 3 Statistical analysis of reddit memes upload times in UTC: coefficients and p-values

Feature	Coefficient	p-Value	Statistically significant
Late night	−0.3879	0.0227	Yes
Early morning	0.0346	0.8611	No
Morning	0.0938	0.6016	No
Late morning	−0.0595	0.7479	No
Afternoon	0.4583	0.003	Yes
Late afternoon	0.3815	0.0055	Yes
Evening	−0.3612	0.0182	Yes
Night	−0.2761	0.0581	Yes

4.1 Image in meme results

Our research into image-related features of memes identifies significant attributes that impact virality, as shown in Fig. 12 (see Appendix C). The top six influential features include Thumbnail Height, Thumbnail Width, Average Value, Average Saturation, Animal Presence in the Image, and Multiple Object Count. These results highlight the importance of visual elements in predicting meme popularity. The magnitude and direction of these features are detailed in Table 1. Among the evaluated models, Model 1, which consists of image features only, represents CatBoost. Its AUC-ROC result of 0.71, depicted in Fig. 10, demonstrates it as the best performer.

4.2 Text in meme results

Our textual analysis of memes identifies crucial features that drive virality, listed in Fig. 13 (see Appendix C). The six main text-related features are Reddit Post Title Word Count, Title Sentiment (Positive and Negative), Submission Text Indicator, Call to Action Indicator, and Memes Text Positive Sentiment. The magnitude and direction of these features are presented in Table 2. Like the image-based analysis, CatBoost proves to be the most effective model, with its 0.73 AUC-ROC result shown in Model 2 in Fig. 10.

Table 4 Assessing AUC-ROC significance with DeLong's test

Comparison	*z*-Value-statistic	*p*-Value
Image-related features AUC-ROC vs. Text-related features AUC-ROC	−0.48615	0.6269
Combined features AUC-ROC vs. Text-related features AUC-ROC	−2.2473	0.0246
Combined features AUC-ROC vs. Image-related features AUC-ROC	−2.9136	0.0036

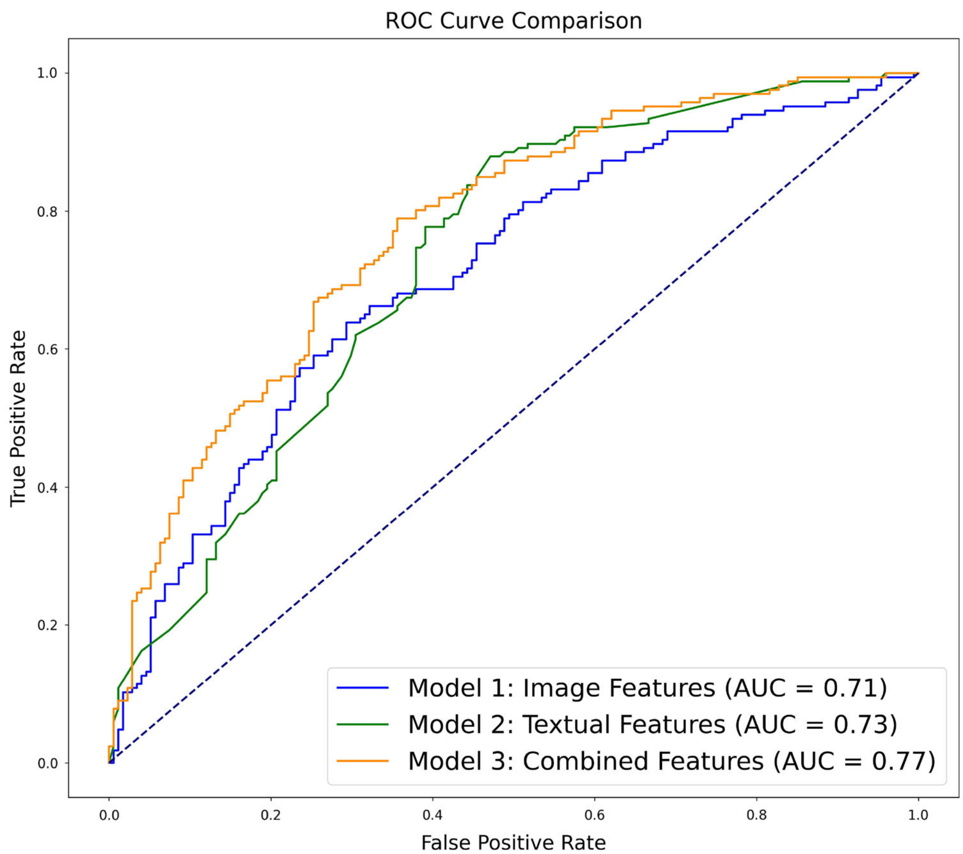


Fig. 10 AUC-ROC of text features, image features, and combined features of memes

4.3 Memes upload times

The upload times of memes, especially during Afternoon, Evening, and Night in UTC, significantly correlate with virality. The analysis suggests that posts in the Afternoon, Late Afternoon, and Evening have a higher chance of going viral, as indicated in the corresponding Table 3.

4.4 Assessing AUC-ROC significance with DeLong's tests

By employing DeLong's tests, we evaluated the statistical significance of AUC-ROC differences between models based on text and image features separately. The findings, documented in a dedicated Table 4, reveal no substantial difference in AUC-ROC values between textual and image features, implying similar contributions to virality. Nevertheless, when these features are combined, the resulting AUC-ROC significantly exceeds those of text or image features alone.

4.5 Combined image and text features for memes

An in-depth examination of memes' combined image and text features shows that Thumbnail Height, Thumbnail Width, Average Value, Average Saturation, Reddit Post Title Word Count, and Title Positive Sentiment are the six key factors in predicting virality as shown in Fig. 11. This integrated analysis, illustrated in Fig. 10, confirms that combining textual and visual elements, as represented by Model 3, leads to a higher AUC-ROC of 0.77 compared to analyzing these features separately.

To determine which combination of attributes made for the most viral factors among image, text, and time-related variables, we used a backward elimination algorithm [31]. The results of this analysis are shown in Table 5.

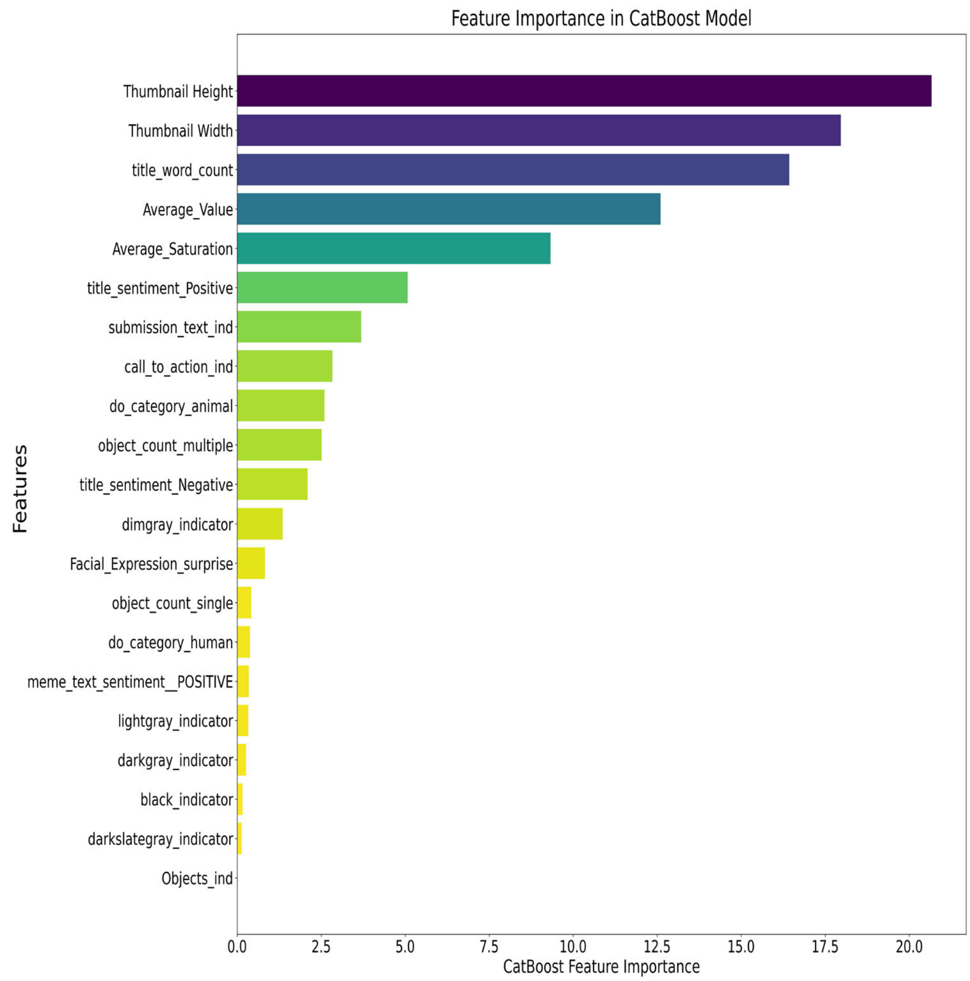


Fig. 11 Combined feature importance of text and image features

5 Discussion

Our analysis reveals that meme virality can be predicted with nearly 80% accuracy using image and text-based features. Including more objects, people, or animals, using bright, low-saturation colors (as measured by higher average brightness and lower saturation), positivity in text and titles, and posting in the early afternoon all predict increased virality of memes. Thumbnail height exhibits a nuanced effect: univariate analysis indicates that decreasing thumbnail height correlates with higher virality, but multivariate analysis reverses this finding, suggesting increasing thumbnail height benefits virality once other variables are controlled. On the other hand, focus on a single object (or person or animal), use of surprised facial expressions, darker, more saturated colors, more associated text (especially negative words and call-to-action words), and posting later in the day predict decreased virality.

Some of our findings diverge from past work in this area. Where Ling et al. reported that both positive and negative emotions boost meme virality, we found surprise decreased virality [16]. This discrepancy may be due to differences in sample size and diversity. On the other hand, our results on average hue, saturation, and value align with Barnes et al.'s study, though average hue did not show a statistically significant relationship in our analysis [17].

From a theoretical perspective, the findings of this study align with principles of Social Cognitive Theory and Uses & Gratifications Theory. Visual features, such as the presence of humans, animals, and lighter colors, highlight the role of attention and retention in driving meme engagement, as suggested by Social Cognitive Theory. Temporal patterns, such as higher virality during afternoon hours, illustrate the influence of contextual reinforcement on user behavior. Additionally, textual features like positive sentiment align with motivations outlined in Uses & Gratifications Theory, such as entertainment and social interaction. While call-to-action phrases may be intended to boost upvotes, our results suggest that these explicit prompts can appear inauthentic, ultimately reducing meme virality. These findings provide insights into how cognitive and motivational factors influence meme virality on Reddit.

Table 5 Statistical analysis of combined features in memes: coefficients and *p*-values

Feature	Coefficient	*p*-Value
Facial expressions surprise	−1.1128	0.008
Negative sentiment in reddit post title	−0.8919	0.000
Presence of animals	0.8040	0.000
Presence of submission text	−0.7823	0.001
Positive sentiment in meme text	0.7653	0.000
Negative sentiment in meme text	0.6466	0.000
Late night	−0.4058	0.019
Presence of single objects	−0.4035	0.001
Night	−0.3745	0.013
Call-to-action indicator in memes	−0.3497	0.007
Evening	−0.3239	0.039
Afternoon	0.3202	0.041
Presence of humans	0.2595	0.020
Average value	0.0037	0.000
Average saturation	−0.0037	0.007
Thumbnail width	−0.0006	0.000
Thumbnail height	0.0003	0.007

From a practical application perspective, marketers and content creators can employ several data-driven strategies to craft engaging meme content that enhances brand visibility and organic engagement without relying on paid promotions. In terms of imagery, using visuals with multiple objects, humans, or animals can make content more relatable and appealing to a broad audience. Avoiding surprise facial expressions is crucial, as they may distract from the intended message or alienate viewers. Opting for less saturated colors and using brighter images enhances visual appeal and ensures that key elements, such as text or logos, stand out. Optimized thumbnail dimensions, particularly smaller and proportional sizes, improve engagement and make content more viral. For text, concise and positive titles are essential to encourage upvotes, while negative sentiment in post titles should be avoided to prevent disengagement. Positive meme text sentiment resonates better with audiences and fosters goodwill, whereas explicit call-to-action phrases like "comment" or "share" should be avoided to maintain authenticity and encourage natural interactions. Timing also plays a critical role in maximizing meme virality. Posts made in the afternoon (12 PM to 3 PM UTC) align with peak online activity across multiple time zones, making this the optimal time for engagement. Conversely, posts made late at night or in the evening tend to underperform.

6 Future directions and limitations

This study relied on data from Reddit, a platform characterized by user anonymity, which may influence the observed relationships. Future research is needed to verify these findings on less anonymous platforms and to ensure their reliability over time through longitudinal studies. Additionally, subreddits with a high volume of posts may disproportionately affect the results, potentially limiting the applicability of insights to smaller subreddits.

Exploration of advanced aesthetic features, such as the Rule of Thirds and visual complexity, could provide deeper insights into the visual elements influencing meme virality. Future studies should also investigate whether textual and image features behave differently across platforms and consider including formats like `.gif` memes to gain a more comprehensive understanding.

This study utilized Python libraries such as OpenCV for image processing, EasyOCR for text recognition, the Hugging Face Transformer library for sentiment analysis, and YOLOv5s for object detection. Advances in these technologies could further enhance the accuracy of such analyses in the future.

7 Conclusions

In conclusion, our study unveils the critical role of both image and textual attributes in determining meme virality on Reddit. By dissecting and analyzing the nuances of these features, we have underscored the complexity behind meme virality and highlighting a multifaceted approach. Notably, our findings reveal that combining textual and visual elements yields a significant predictive advantage, marking a substantial leap over traditional, one-dimensional analyses.

For data scientists, natural language processing, and computer vision researchers, our study provides a foundational basis for developing more accurate predictive models by incorporating the identified influential features. Additionally, content moderators could benefit from prioritizing the review of content that possesses viral features, thereby streamlining their moderation efforts. Furthermore, educators and communicators can use our research to make presentations and materials more engaging. Understanding what makes meme go viral helps in creating attention-grabbing and shareable materials. This boosts the spread and impact of educational and informational content.

Overall, our research enhances the comprehension of virality among Reddit social media memes, providing valuable implications for academic research and practical applications in the digital landscape. Furthermore, our GitHub repository hosts our code for extracting and analyzing meme features and serves as a valuable resource for further research. Additionally, 16,968 memes extracted from Reddit are stored on the Open Science Framework (OSF), encouraging the community to build upon our work, enhance predictive models, and gain deeper insights into meme virality.

Appendix A: Extracted and filtered posts from selected subreddits

Table 6 represents extracted and filtered posts from selected subreddits in January 2024.

Table 6 Extracted and filtered posts from selected subreddits in January 2024

Subreddit	Subscriber count	Number of extracted posts	Number of filtered posts
youdontsurf	382k	808	808
oddlysatisfying	10.4m	88	88
interestingasfuck	11.7m	148	147
ExpectationVsReality	2 m	276	276
AntiMeme	956k	814	812
lostredditors	776k	760	760
woooosh	1.1m	268	268
absolutelynotme_irl	507k	854	853
2meirl4meirl	1.5m	854	854
sadcringe	1.3m	267	267
blackpeopletwitter	5.7m	451	451
WhitePeopleTwitter	3.1m	812	810
PoliticalHumor	1.6m	836	836
DeepFried	18.5k	871	871
ComedyCemetery	1 m	435	435
AdviceAnimals	10 m	407	407
PrequelMemes	2.8m	733	733
ProgrammerHumor	3.4m	594	594
HistoryMemes	9.1m	641	641
terriblefacebookmemes	2.2m	737	737
JustUnsubbed	234k	617	617
surrealmemes	904k	391	391
bonehurtingjuice	903k	596	595
dank_meme	785k	688	688
me_irl	7.6m	662	662
memes	29.4m	348	348
wholesomemes	714k	757	757
dankmemes	6 m	670	670
wholesomememes	15.2m	585	585

Appendix B: Extracted and engineered features from reddit posts

Table 7 presents a comprehensive overview of both the extracted raw features and the subsequently engineered features derived from Reddit posts.

Table 7 Extracted and engineered features from reddit posts

Column Name	Description
Subreddit	The name of the subreddit where the post was submitted
Title	The title of the submitted post
Upvotes	The number of upvotes received by the post
UTC	The Coordinated Universal Time (UTC) when the post was created
Image URL	The URL pointing to the image in the post
Filename	The name of the file associated with the image
NSFW	A boolean indicating whether the post is marked as "Not Safe For Work" (NSFW)
Num comments	The number of comments received on the post
Author	The username of the author who submitted the post
Submission ID	The unique identifier for the post submission
URL Domain	The domain extracted from the URL of the submission
Upvote Ratio	The ratio of upvotes to total votes received by the post
Post Flair	Text indicating any flair assigned to the post
Post Permalink	The permanent link to the post
Submission Text	The text content provided in the post
Thumbnail Height	The height of the thumbnail associated with the post
Thumbnail Width	The width of the thumbnail associated with the post
Has Thumbnail	A boolean indicating whether the post has an associated thumbnail
Submission Date	The date the meme was posted
Top_Color	The most dominant color in the image, in RGB format
Top_Color_Name	The name of the most dominant color in the image
Average_Hue	The average hue value from the image
Average_Saturation	The average saturation value from the image
Average_Value	The average value/brightness from the image
Average_Red/Green/Blue	The average values for red, green, and blue channels in the image
Hex_Color	The hexadecimal color code of the dominant color
Hue/Saturation/Value	Hue, Saturation, Value components for the image
Red/Green/Blue	Red, Green, Blue components of a color for the image
Hex_Color	The hexadecimal color code of the dominant color

Table 7 continued

Column Name	Description
Hue/Saturation/Value	Hue/Saturation/Value for the image
Red/Green/Blue	Red/Green/Blue component of a color for the image
Detected Objects	The objects detected in the meme image
Extracted_Text	Text extracted from the image of the meme
Expression_ind	An index or indication of facial expressions found in the image
Objects_ind	An index or indication of objects found in the image
object_count_multiple/no object/single	Counts of objects in the image categorized as multiple, none, or single
do_category	Categories of detected objects in the image
do_category_Other	Categories for objects in the image that do not fit into predefined groups
do_category_accessory	Identifies if the image contains accessories (like jewelry, hats, etc.)
do_category_animal	Identifies if the image contains animals
do_category_appliance	Identifies if the image contains household appliances
do_category_ball	Identifies if the image contains ball-shaped objects
do_category_clothing	Identifies if the image contains items of clothing
do_category_container	Identifies if the image contains containers (like boxes, bottles, etc.)
do_category_device	Identifies if the image contains electronic devices
do_category_food	Identifies if the image contains food items
do_category_furniture	Identifies if the image contains furniture
do_category_human	Identifies if the image contains humans or human figures
do_category_object	General category for any objects in the image
do_category_plant	Identifies if the image contains plants
do_category_sports equipment	Identifies if the image contains sports equipment
do_category_tool	Identifies if the image contains tools
do_category_toy	Identifies if the image contains toys
do_category_vehicle	Identifies if the image contains vehicles
Facial_Expression_angry	Indicates the presence of an angry facial expression in the image
Facial_Expression_disgust	Indicates the presence of a disgusted facial expression in the image
Facial_Expression_fear	Indicates the presence of a fearful facial expression in the image
Facial_Expression_happy	Indicates the presence of a happy facial expression in the image
Facial_Expression_neutral	Indicates the presence of a neutral facial expression in the image

Table 7 continued

Column Name	Description
Facial_Expression_sad	Indicates the presence of a sad facial expression in the image
Facial_Expression_surprise	Indicates the presence of a surprised facial expression in the image
Facial_Expression_unknown	Indicates that the facial expression could not be determined
meme_text_sentiment_NEGATIVE	Identifies whether the sentiment of the text in the meme is negative
meme_text_sentiment_NEUTRAL	Identifies whether the sentiment of the text in the meme is neutral
meme_text_sentiment_POSITIVE	Identifies whether the sentiment of the text in the meme is positive
meme_text_sentiment_Unknown	Identifies whether the sentiment of the text in the meme is unknown
Time_12am-3am	The meme was posted between midnight and 3 am
Time_12pm-3pm	The meme was posted between noon and 3 pm
Time_3am-6am	The meme was posted between 3 am and 6 am
Time_3pm-6pm	The meme was posted between 3 pm and 6 pm
Time_6am-9am	The meme was posted between 6 am and 9 am
Time_6pm-9pm	The meme was posted between 6 pm and 9 pm
Time_9am-12pm	The meme was posted between 9 am and noon
Time_9pm-12am	The meme was posted between 9 pm and midnight
call_to_action_ind	Indicates if the meme has a call-to-action
submission_text_ind	Indicates if there is text accompanying the meme submission
NSFW_ind	Indicates if the meme is not safe for work (contains explicit content)
title_word_count	The number of words in the meme's title
title_sentiment	The overall sentiment of the meme's title
title_sentiment_Negative	Indicates if the sentiment of the title is negative
title_sentiment_Positive	Indicates if the sentiment of the title is positive
Height/Width Ratio	The ratio of the meme image's thumbnail height to its thumbnail width
Width*Height Product	The product of the meme image's thumbnail width and thumbnail height
Normalized_Upvotes	The number of upvotes normalized using min max scaler

Table 7 continued

Column Name	Description
Ventile	It represents a value indicating the distribution of upvotes for a particular subreddit, where a Ventile of 20 corresponds to the top 5th percentile of upvotes, signifying the highest engagement within that subreddit. Conversely, a Ventile of 1 represents the bottom 5th percentile of upvotes, indicating the lowest engagement. This numerical scale continues accordingly for each Ventile value in between
gray_indicator	Represents whether this color is the dominant color in memes or not
darkslategray_indicator	Represents whether this color is the dominant color in memes or not
dimgray_indicator	Represents whether this color is the dominant color in memes or not
darkgray_indicator	Represents whether this color is the dominant color in memes or not
silver_indicator	Represents whether this color is the dominant color in memes or not
black_indicator	Represents whether this color is the dominant color in memes or not
rosybrown_indicator	Represents whether this color is the dominant color in memes or not
lightgray_indicator	Represents whether this color is the dominant color in memes or not
gainsboro_indicator	Represents whether this color is the dominant color in memes or not
whitesmoke_indicator	Represents whether this color is the dominant color in memes or not

Appendix C: Feature importance of text and image features

See Figs. 12 and 13

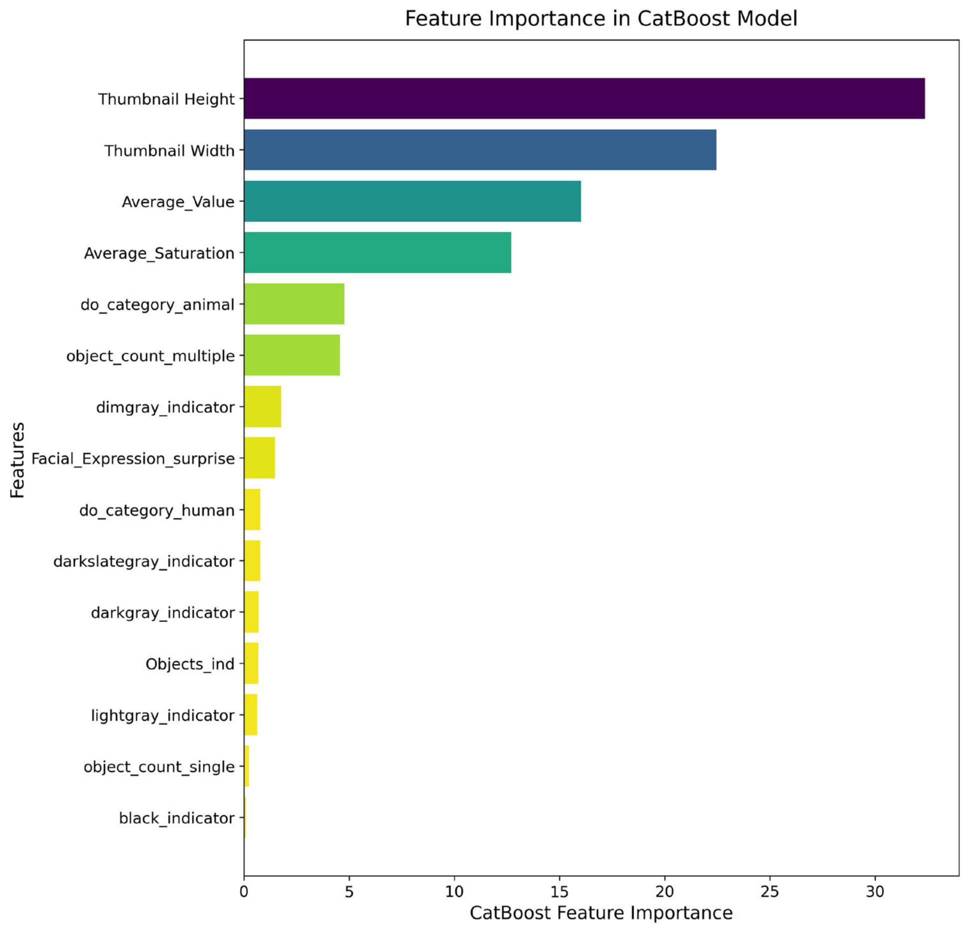


Fig. 12 Feature importance of image features

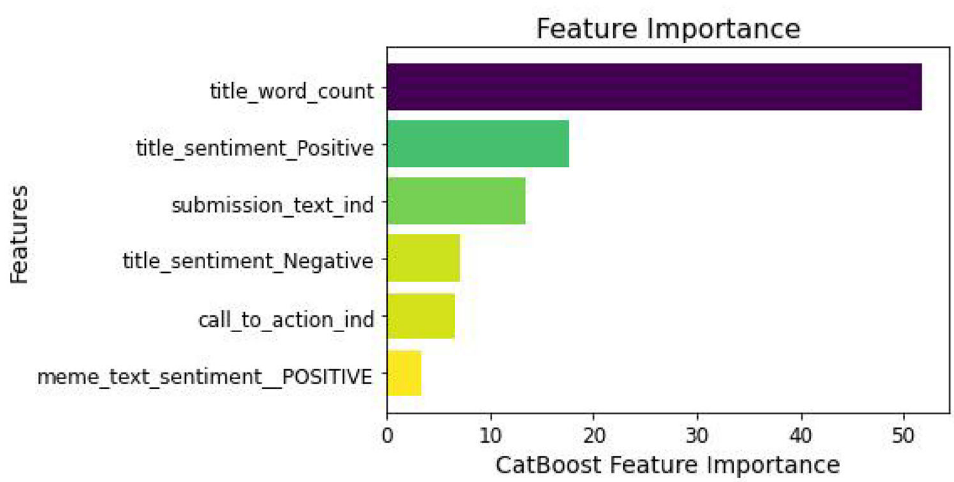


Fig. 13 Feature importance of text features

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Data availability The data are available at the Open Science Framework repository: https://doi.org/10.17605/OSF.IO/PV9MY. The repository features viral, non-viral, and memes of intermediate popularity, in addition to the Python code and CSV files required to replicate the study.

Code availability The code to replicate this study is available on GitHub at https://github.com/tradertanmay/reddit-memes-virality-prediction.

Declarations

Conflict of interest The authors have no conflict of interest.

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